

**Rhetoric vs. Reality: A Comparative Analysis of Global Space Sustainability Definitions
and Compliance**

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11 January 2026



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Introduction

It is difficult to quantify the vastness of Earth's orbit definitively, yet it is a finite resource. Once perceived as a boundless frontier, the orbital environment has become a congested, competitive domain that everyone and every nation must share. Space debris is the biggest threat to orbital stability in the future. Throughout this paper, the elimination of the threat of orbital debris and space sustainability will be closely linked, but not identical in description. Some definitions of space sustainability focus on clearing and preventing space debris, while others may mention accessibility or 'resource' sharing. A clean orbit is necessary for future satellites and missions. It is vital to a nation's security, economy, and communication, which is why there needs to be a unified definition to guide the world toward a sustainable orbit.

The global community has sought to define a sustainable use of outer space. The United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) established a baseline with a definition. Most space-faring nations in the United Nations (UN) have adopted this UNCOPUOS definition regarding space and orbital sustainability.

A unified definition of the necessity of space sustainability does not exist; definitions vary widely by national interest. These definitions fragment into three competing logics: environmental preservation, security dominance, and equitable development. Furthermore, a significant compliance gap (the rhetoric vs. reality gap) exists, in which voluntary guidelines are frequently superseded by national security and commercial/economic priorities. For a more in-depth understanding of these divergences in definition, this literature review categorizes these approaches into four coherent groups: The Multilateral Normative Baseline, The Strategic Realists, The Technical Stewards, and The Developmental Bloc.

The Normative Baseline

Multilateral definitions serve as the diplomatic "lowest common denominator," establishing broad norms rather than enforceable laws. These definitions are primarily negotiated within the United Nations and technical standard-setting bodies to provide a shared vocabulary for diverse actors.

The United Nations Framework

The governance of space has undergone a significant paradigm shift from the hard-law era of the 1967 Outer Space Treaty to the current soft-law approach, as reflected in the 2019 Long-Term Sustainability (LTS) Guidelines. According to the Guidelines for the Long-term Sustainability of Outer Space Activities (UNOOSA, n.d.), the UN defines sustainability as a cross-sectoral issue involving safety, security, and development. The definition promotes environmental preservation, equitable access, and the peaceful use of space (UNOOSA, n.d.). The COPUOS LTS definition is as follows:

"The long-term sustainability of outer space activities is defined as the ability to maintain the conduct of space activities indefinitely into the future in a manner that realizes the

objectives of equitable access to the benefits of the exploration and use of outer space for peaceful purposes, to meet the needs of the present generations, while preserving the outer space environment for future generations. This is consistent with, and supports, the objectives of the Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space and the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (Outer Space Treaty), as such objectives are integrally associated with a commitment to conducting space activities in a manner that addresses the basic need to ensure that the environment in outer space remains suitable for exploration and use by current and future generations. States understand that maintaining exploration and use of outer space for peaceful purposes is a goal to be pursued in the interest of all humankind.” (United Nations [UN], 2019)

The framework consists of 21 voluntary guidelines that address the policy and regulatory environment, the safety of space operations, scientific and technical research, and international cooperation. A core element of the UN definition is its alignment with the UN Sustainable Development Goals (SDGs), particularly regarding Space for Sustainability on Earth. The UN views space sustainability as closely linked to terrestrial resilience, with orbital assets supporting climate action and disaster relief. The primary limitation of the UN framework is its voluntary, non-binding nature.

The Technical Baseline

Technical bodies sharpen the definition of sustainability through engineering and physical metrics. The Inter-Agency Space Debris Coordination Committee (IADC) provides the most authoritative technical definition of space debris and sustainability:

"Any activity that takes place in outer space should be performed while recognising the unique nature of the following regions... to ensure their future safe and sustainable use. These regions shall be protected regions with regard to the generation of space debris." (Inter-Agency Space Debris Coordination Committee [IADC], 2021)

Sustainability, from the IADC perspective, is defined through debris mitigation. This includes the 25-year rule for post-mission disposal, which requires satellites in Low Earth Orbit (LEO) to re-enter the atmosphere or be moved to a graveyard orbit within 25 years of mission completion (Inter-Agency Space Debris Coordination Committee [IADC], 2021). However, recent literature, such as the Space Sustainability Rating (SSR), notes that this rule is being outpaced by the sheer volume of new launches (European Space Agency [ESA], 2024). This has led to calls for a stricter 5-year limit to preserve the carrying capacity of the LEO and GEO regions, which is the maximum number of objects an orbit can safely hold without triggering a runaway debris chain reaction (D’Amico et al., 2025; ESA, 2024).

Similarly, the International Organization for Standardization (ISO), through standards such as ISO 24113, translates these goals into verifiable engineering requirements:

"The primary objective of this document is to reduce the growth of space debris by ensuring that space systems are designed, operated and disposed of in a manner that prevents them from generating debris throughout their orbital lifetime. Implementation of this document is a key element in ensuring the long-term sustainability of outer space activities." (International Organization for Standardization [ISO], 2023).

In this context, sustainability is defined as design for demise, which is the technical capability of a spacecraft to disintegrate during re-entry to minimize risk to people on Earth. The ISO has a specific 90% threshold that many national policies endorse as a non-legally binding method. This requires a 90% success rate for post-mission disposal.

The Spectrum Resource

Finally, the International Telecommunication Union (ITU) provides a unique dimension to the definition:

"In using frequency bands for radio services, Member States shall bear in mind that radio frequencies and any associated orbits, including the geostationary-satellite orbit, are limited natural resources and that they must be used rationally, efficiently and economically, in conformity with the provisions of the Radio Regulations, so that countries or groups of countries may have equitable access to those orbits and frequencies..." (International Telecommunication Union [ITU], 2011).

The ITU manages sustainability through rational use and equitable access. It established that space is not just a vacuum but also a limited resource, specifically radio frequencies and orbital slots. These 'resources' must be managed with a focus on efficiency and equity. The ITU addresses how developing countries will have a more difficult time accessing this orbital resource and are effectively barred from orbit by first movers like the U.S. and China. However, the ITU favors those able to launch quickly. A single company or nation can occupy a frequency or orbital position, making it difficult for a smaller nation to find a slot for its own sovereign satellite.

Many of these definitions provide the reality of the situation. There are high expectations for these guidelines and definitions, which require strict adherence to mitigating space debris. National governments are the entities responsible for what this paper will deem rhetoric when they provide their own definitions of space sustainability.

The Strategic Realists

Strategic Realists represent the traditional and dominant spacefaring powers: the United States of America, the Russian Federation, and the People's Republic of China. To these

nations, sustainability is not an environmental goal but a necessity for maintaining national security, strategic deterrence, and geopolitical hegemony. These countries have the most assets in space, whether strategic, military, or commercial; they have invested heavily in this environment. This will undoubtedly lead these nations to act in their own best interests, especially if they have much to lose.

The United States: Sustainability as Operational Stability

"The United States considers the sustainability, stability, and use of space vital to its national interests. ... The increasingly congested and contested nature of the space environment offers critical challenges, which threaten the long-term sustainability of our space activities" (U.S. Department of State, 2012). This approach is primarily focused on protecting high-value assets and ensuring that commercial leadership can thrive without the risk of catastrophic disruption. A more recent definition from 2024 states that space sustainability is "The ability to maintain the conduct of space activities indefinitely into the future in a manner that is safe, peaceful, and responsible to meet the needs of the present generations while preserving the outer space environment for future activities and limiting harm to terrestrial life" (NASA, 2024). This rhetoric is NASA seeking a safer, more responsible role. On the other hand, The White House (2020) mentions global cooperation, environmental preservation, and legal resource extraction. However, it also prioritizes commercial leadership, space as a warfare domain, and freedom of operation (The White House, 2020).

As analyzed by D'Amico et al. (2025), a critical evolution in the American approach to orbital health is the reallocation of Space Traffic Management (STM) from military to civilian control. This move signals a shift toward a more transparent, civil-led model of sustainability (2025). The core of this transition is the Office of Space Commerce (2024) Traffic Coordination System for Space, a modern architecture designed to provide basic space situational awareness (SSA) services and collision warnings to civil and private operators (Office of Space Commerce, 2024).

By providing open-source tracking data, the U.S. aims to set global standards for responsible behavior that align with its commercial interests. This shift allows the Department of Defense to focus on core military space domain awareness. At the same time, the Department of Commerce manages the safety of flight missions for the rapidly expanding commercial sector.

The United States demonstrates a significant gap between rhetoric and reality. On one hand, it spearheads the Artemis Accords, a diplomatic framework that promotes safety zones and "mitigation of orbital debris" (NASA, 2024) to ensure the safe and sustainable exploration of space. Its domestic regulatory landscape reveals a contradictory reality. While the FCC (2022) has introduced a strict 5-year deorbit rule (FCC-22-74), requiring satellite operators to dispose of their spacecraft shortly after mission completion, the agency continues to grant licenses towards thousands of satellites for commercial mega-constellations (FCC, 2022).

Critics argue that by authorizing tens of thousands of satellites, the U.S. is effectively "occupying the carrying capacity of LEO" (Suess, 2023). This creates a tension in equity where

the U.S. uses the rhetoric of sustainability to lock in a first-mover advantage while physically creating the debris it warns against. The United States shows how a nation can have multiple definitions of space sustainability across different agencies. They have definitions that often shift depending on which administration is in power. This internal inconsistency represents the most significant gap among all spacefaring nations.

The Russian Federation: Sustainability as Sovereignty

The Russian Federation defines sustainability through the lens of legal purism, often prioritizing state sovereignty and the prevention of militarization over purely technical debris mitigation. As seen in the Permanent Mission of the Russian Federation to the United Nations (2021), its definition of sustainability is closely linked to the Prevention of an Arms Race in Outer Space (PAROS). The sustainability of outer space activities, both in the short and long term, requires the international community to ensure that no weapons are ever placed or used there, as part of a broader commitment to preserving outer space as a peaceful environment (Permanent Mission of the Russian Federation to the United Nations, 2021). For Moscow, an environment filled with orbital weapons, including those potentially disguised as active debris-removal (ADR) tools, is inherently unsustainable, rendering technical safety measures meaningless without a legally binding security treaty.

Russia presents one of the most stark contradictions between rhetoric and reality. While diplomatically advocating for the "long-term sustainability of outer space activities" (UNCOPUOS, 2019). According to U.S. Space Command (2021), Russia conducted a destructive direct-ascent anti-satellite (ASAT) test in November 2021 against its defunct Cosmos 1408 satellite. This event produced over 1,500 pieces of trackable debris and millions of smaller fragments (U.S. Space Command, 2021), which crossed the International Space Station's orbit. This directly contradicts its stated commitment to the IADC (2021) on the UN Guideline B.4 regarding the performance of conjunction assessments and the minimization of intentional debris (IADC, 2021). This highlights a reality where environmental sustainability is frequently subordinated to military signaling and kinetic deterrence.

The People's Republic of China: Sustainability as High-Quality Development

China's definition of sustainability has evolved under the concept of a "Shared Future for Mankind" (Khan et al., 2021). In its strategic outlook, China defines sustainability as a domain for socioeconomic development and national defense, emphasizing the responsible use of space resources and the modernization of space governance. According to the State Council (2022), China views its space program as a vehicle for national rejuvenation, aiming to protect the outer space environment while simultaneously leveraging orbital assets to enhance national security and high-quality economic development (State Council, 2022).

While China positions itself as a steward of the Global South within the IADC and UNCOPUOS, its operational history reveals a significant gap between rhetoric and reality. According to Kelso (2007), China's 2007 ASAT test remains the single largest debris-creating event in history, fundamentally contradicting the shared future narrative (Kelso, 2007). The current plans for the Guowang constellation (a state-led network aiming for approximately 13,000 satellites) suggest that China's definition of a shared future may actually prioritize its own national sovereign constellations over the collective carrying capacity of the orbit. As argued in the CSIS Strategic Trajectories Report (2025), this expansion creates an equity tension in which China's rapid occupation of (LEO) effectively limits the equitable access it publicly advocates in international forums (CSIS, 2025).

The Technical Stewards

Technical Stewards treat space sustainability as an environmental and technical imperative, moving the conversation from mitigation (reducing harm) to remediation (fixing the environment). These actors closely follow the guidance and definitions from the multilateral organizations. They set the example when it comes to the rhetoric vs. reality gap, for there are small gaps that show; gaps, nonetheless, but the least mentionable of the lot.

Europe

The European Space Agency (ESA) provides the most aggressive environmental definition of sustainability. Through the Zero Debris Charter, ESA (2023) aims for debris neutrality by 2030, establishing a framework to minimize the impact of human-made objects in the orbital environment (ESA, 2023). Within this framework, ESA (2024) has defined sustainability as a technical requirement where missions must maintain a probability of successful disposal exceeding 90 percent and utilize a design for demise to ensure satellites burn up completely upon re-entry (ESA, 2024).

Europe sticks to its word when it comes to the gap between rhetoric and reality. ESA (2024) is currently pioneering the ClearSpace-1 mission, the first active debris removal operation to target and de-orbit legacy objects (ESA, 2024). By integrating ISO 24113 standards into its procurement contracts, the European Union uses its economic power to enforce compliance among commercial partners. This transitions sustainability from a voluntary goal into a mandatory industrial standard, effectively bridging the rhetoric vs. reality gap through contractual accountability.

Japan

According to Montluc (2011), Japan views sustainability through the lens of stable use and technological breakthroughs. By integrating debris mitigation into its National Security Strategy, Japan has effectively linked the future health of the orbital environment to its own national security interests (Montluc, 2011). The Japan Aerospace Exploration Agency (JAXA) (2022) made a clear commitment to preserving outer space as a realm for all of humanity,

ensuring that humanity can continue to benefit from it for future generations (Japan Aerospace Exploration Agency, 2022).

Japan is a global leader in turning sustainability rhetoric into hardware. According to Xu (2019), regarding orbital robots, JAXA is partnering with private firms such as Astroscale to actively target and remove large-scale legacy debris through the Commercial Removal of Debris Demonstration mission. In 2024, the ADRAS-J mission successfully demonstrated the world's first safe approach and inspection of a non-cooperative rocket upper stage using advanced Rendezvous and Proximity Operations (Xu, et al., 2019). By establishing specific safety requirements for these missions, Japan is defining sustainability through a trust-but-verify model of space situational awareness that prioritizes transparency and responsible behavior.

South Korea

South Korea has emerged as a key Technical Steward by integrating sustainability as a space power. In its formal address to the UN, the Republic of Korea emphasized the need of “ensuring a sustainable future for our planet through space technology,” (Republic of Korea, 2025b), mainly focusing on the improvement of satellite data to combat climate change and ...” maintain an operationally stable and safe environment for the space economy” (Republic of Korea, 2025a).

South Korea has actively adopted the UN LTS Guidelines into its national space law. In practice, the Korea Aerospace Research Institute has focused on developing autonomous collision avoidance systems and has participated in international debris-tracking networks. By coordinating its technological roadmap with sustainable orbit objectives, South Korea seeks to avoid the rhetoric-versus-reality gap common among larger powers, positioning sustainability as a prerequisite for its national satellite operations.

The Developmental Bloc

For emerging space nations, sustainability is defined by equity: the right to access space before it is effectively closed off by debris or regulations imposed by established powers. These players have contributed the least to the formation of space debris in Earth's orbit. Granted, it is because they do not have the same level of access as established space actors.

Again, emerging space nations do not hold the same privileges to the orbital commons as established powers. As noted by Jakhu and Pelton (2017), the current trajectory of orbital deployment suggests a future of orbital foreclosure, where limited spatial capacity makes it virtually impossible for emerging agencies to operate safely (Jakhu & Pelton, 2017). This creates a permanent rhetoric-versus-reality gap: while international law guarantees space as a place for all of humanity, the physical reality of congestion prevents emerging states from exercising their technological sovereignty in the new space economy.

AfSA and ALCE

The African Space Agency (AfSA) and the Latin American and Caribbean Space Agency (ALCE) interpret sustainability through the lens of socioeconomic development and equitable access. In these regions, sustainable spaces can be used to address terrestrial challenges such as climate change, agriculture, and disaster management. The space sector is a means to grow and maintain sustainability on Earth.

The African Union (2023) says that AfSA is tasked with acquiring space technologies for sustainable development and fostering an independent space sector responsive to socioeconomic challenges, while suggesting that major space-faring nations should fulfill their responsibility to mitigate space debris (African Union, 2023). This means the AfSA should do its best to acquire data and technology to get them into orbit, while developing its private sector. Furthermore, they demand that space be made for them in Earth's orbit. The Latin American and Caribbean Space Agency (2021) coordinates efforts to monitor infrastructure and environmental health from space, ensuring that member nations have the capacity to track changes that affect the environment, while boosting the peaceful use of outer space for the region's development (ALCE, 2021). The ALCE focuses on Earth observation data and ways that the space sector can directly benefit them.

These actors, and other emerging space organizations/nations, argue that the strict technical standards proposed by the Technical Stewards act as soft-law barriers. Soft law limits Earth's orbital space by imposing technological barriers that emerging nations cannot overcome. According to Froehlich (2019), if emerging nations are required to use expensive ADR-compatible technology or high-precision tracking, they may be effectively priced out of the market (Froehlich, 2019). Space technology is expensive and beyond the scope of nations that cannot afford it. This creates an equity tension that emerging nations view as "technological colonialism...[and] first-come, first-served" (Wood & Weigel, 2012). Emerging nations treat the nature of orbital allocation as a threat to their future ability to participate in the space economy. True sustainability is defined here as a regime that includes capacity building and technology transfer.

Critical Analysis

The academic literature reveals that while the vocabulary of space sustainability has been standardized, the mechanisms for ensuring compliance are profoundly underdeveloped. There is a saying: Do not trust what governments say; watch what they do. The rhetoric nations usually release should always be taken with a grain of salt, for the reality of their actions will likely do the exact opposite of what they claim to support. This is the gap. Below is a table that provides examples of this compliance gap.

Nation / Entity	Rhetoric (Diplomatic Commitment)	Reality (Regulatory & Operational Action)	The "Gap" Description
USA	Artemis Accords: Safe, stable, and operational environment.	FCC 5-year deorbit rule vs. licensing of 30,000+ mega-constellation satellites.	Carrying Capacity Gap: Promoting mitigation while occupying the majority of LEO slots.
Russia	UN COPUOS: Prevention of an Arms Race in Outer Space (PAROS).	2021 Nudol ASAT Test (Cosmos 1408) created 1,500+ trackable fragments.	Security-Sustainability Paradox: High-level peace rhetoric vs. kinetic debris creation.
China	2021 White Paper: Community with a shared future for mankind.	2007 ASAT event; current plans for ~13,000 "Guowang" satellites.	Equitable Development Gap: Balancing global shared future with national sovereign constellations.
ESA	Zero Debris Charter: "Zero debris by 2030."	Mandatory 90%+ disposal success rate in all new procurement contracts.	Implementation Alignment: Low gap; rhetoric is codified into binding industrial standards.
IADC / ISO	Technical Norms: 25-year rule and Design for Demise.	Global PMD (Post-Mission Disposal) success rate remains below 60% for many regions.	Compliance Gap: International technical guidelines are voluntary and often ignored due to cost.

Table 1: Global Governance Gaps. Sorted by nation/entity, rhetoric of definition, reality of actions, and gap description; Referenced information in sources.

Three primary gaps define the current state of global governance regarding space sustainability.

The "Voluntary" Trap and the Limits of Soft Law

As Martinez (2019) analyzed, the shift from hard law (treaties) to soft law (LTS Guidelines) has created a regulatory vacuum. Because the 21 LTS Guidelines are voluntary, nations are free to interpret sustainability in ways that prioritize sovereign interests. This leads to a fragmented landscape in which Strategic Realists can conduct kinetic ASAT tests that create thousands of debris fragments while simultaneously claiming diplomatic adherence to sustainability in UN forums. Without a central enforcement mechanism or a 'Space Traffic Police' (Martinez, 2019), the burden of compliance falls on national licensing authorities, which often prioritize domestic economic competition over global environmental goals.

Sustainability Paradox

A significant, emerging gap identified by Jeff Foust (2024) is the neglect of the Earth-space continuum. Current definitions of sustainability focus almost exclusively on clearing the orbit. The standard solution for Post-Mission Disposal is to have satellites burn up in the atmosphere. However, as mega-constellations like Starlink begin to de-orbit thousands of satellites, the literature warns of the injection of alumina particles and soot into the upper stratosphere (Foust, 2024). This unregulated pollution represents a compliance gap in environmental logic: we are protecting the vacuum of space by potentially damaging the delicate chemistry of Earth's atmosphere. This will raise valid questions about the priorities surrounding the immediate sustainability of Earth's environment versus the sustainable use of Earth's orbit.

The Role of the Private Sector

Commercial actors are currently moving at a velocity that far exceeds national, and especially international, regulators. As seen in the Space Sustainability Rating framework, industry-led initiatives are attempting to fill the governance void. According to Rathnasabapathy (2022), the SSR provides a market-based incentive by scoring missions on Detectability, Identifiability, and Trackability (Rathnasabapathy, et al., 2022). This suggests that the definition of sustainability is increasingly being shaped by corporate incentives, social responsibility, and insurance requirements rather than international law. As argued by Löfqvist (2023), the future of compliance may lie in terrestrial Environmental, Social, and Governance (ESG) principles, where private capital (not just state diplomacy) enforces sustainable behavior (Löfqvist, 2023). Commercial actors are looking more into SSR to lower their insurance premiums and attract ESG investors. This creates normal regulation that moves faster than treaty law.

Conclusion

The comparative analysis of global space sustainability frameworks reveals that while a shared vocabulary of safety, stability, and longevity has emerged, the underlying strategic intents remain profoundly divided. The Multilateral Normative Baseline serves as a diplomatic ideal, yet it functions more as a mirror of national priorities than a binding roadmap for action.

The Strategic Realists have effectively co-opted the rhetoric of sustainability to secure "freedom of operation" and national hegemony. Their definitions are security-first, prioritizing the ability to monitor adversaries under the guise of debris mitigation. The Technical Stewards have adopted the most rigorous definitions, treating space as a fragile ecosystem and pioneering the technologies needed to achieve their goal of zero debris. The Developmental Bloc continues to fight for a definition rooted in equitable access, but its gap is not rooted in hypocrisy; it is in capacity.

A compliance disconnect exists between international rhetoric and domestic enforcement. While states publicly advocate for a shared future in outer space, their national regulatory frameworks lack the binding technical requirements necessary to keep commercial

and governmental actors accountable for debris mitigation. ASAT tests and the unregulated expansion of mega-constellations exemplify this. This proves that space sustainability is currently a contested concept. The physical reality of an exponentially polluted orbit does not negotiate with policy ambiguities; every collision in orbit increases the probability of an inevitable failure that would make the orbital commons unusable for all nations.

The literature suggests that space sustainability cannot remain a voluntary endeavor. To bridge the gap between rhetoric and reality, the international community must move toward a harmonized, enforceable regulatory regime. This regime must treat the orbital environment as a finite global commons, ensuring that the needs of the present do not permanently foreclose future celestial opportunities. There is a need for a more coherent and enforceable space sustainability rating system, which could prove to be a more effective tool for pursuing space sustainability than voluntary guidelines.

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